- **Environment:** Build a simplified simulation of a single intersection, where vehicles arrive from different directions and wait in queues. You can begin with two traffic light phases and simple vehicle arrival patterns.
- **RL agent:** Train a tabular Q-learning agent to select traffic light phases. The state could include queue lengths and the current light phase, while the reward could penalise waiting time or long queues. Tabular Q-learning stores action values in a table and does not require a neural network, so it provides a relatively accessible starting point. If your group is interested in deep reinforcement learning, you could optionally extend the project to a Deep Q-Network (DQN) and compare the two approaches.
- **Baselines:** Compare your agent with a fixed-time controller that switches lights on a regular schedule and a queue-based controller that prioritises the direction with more waiting vehicles.
- **Evaluation:** Test under different traffic conditions, such as light traffic, heavy traffic, or uneven flows across directions. Average waiting time and queue length are useful metrics.

Please treat this outline as a reference rather than a rigid specification. Your group is welcome to explore different simulation settings, state and reward designs, or RL methods. Deep reinforcement learning is an optional extension, not a requirement. The main aim is to develop a manageable implementation, make clear comparisons, and analyse what works and why.If you have any other questions, please feel free to contact me.